From **Data** to **Requirements** with Data Analysis

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2025-09-24



Project Work – So Far

- Form groups of 3-5 students.
- Come up with a project idea. What problem or need are you going to solve?
- Think about who your target group is!
- Performed data collection (interview/observation/survey)
- Data in the form of, for example, notes from interviews/observations or survey responses.

Timeline

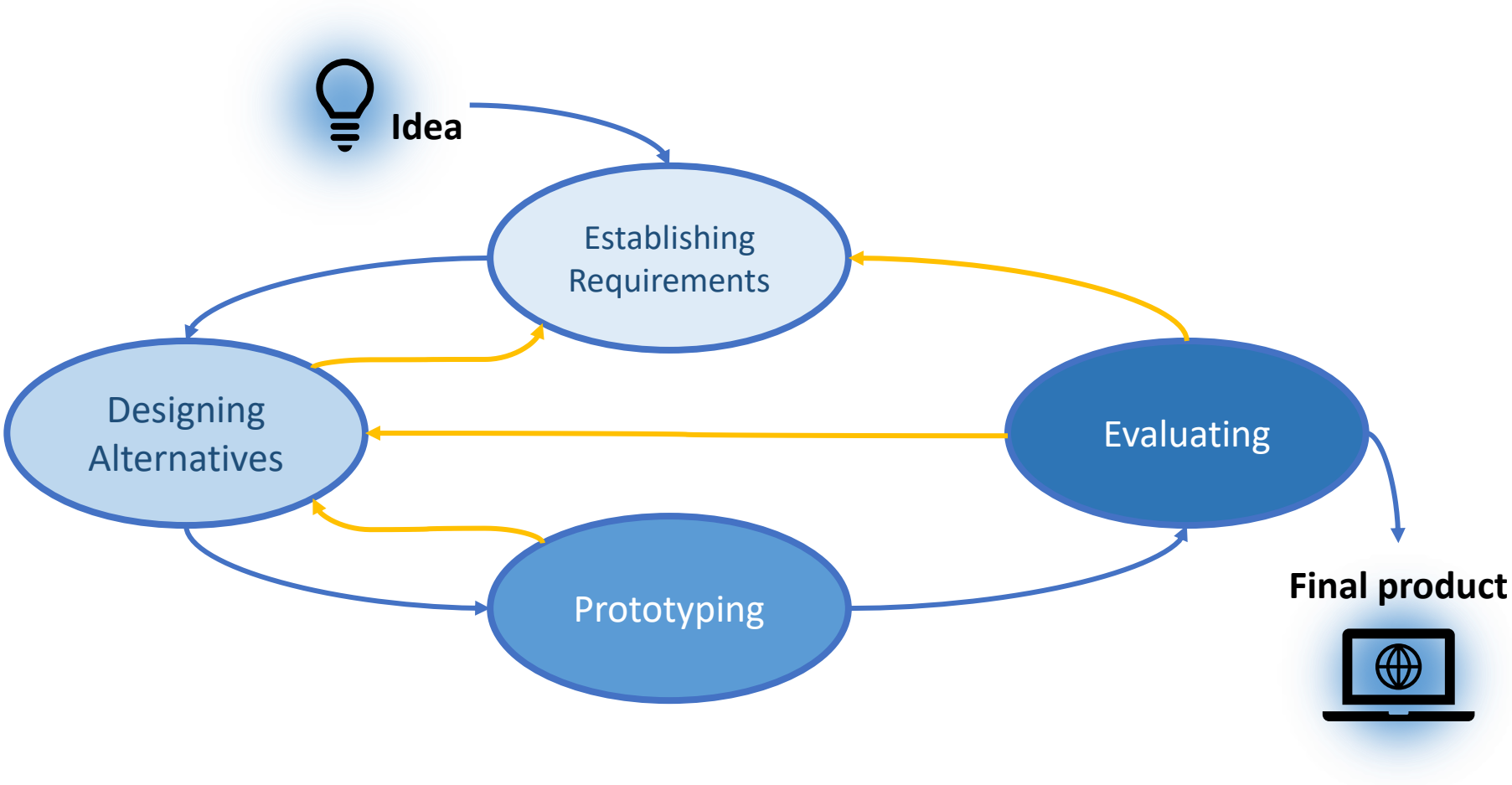
	Mon	Tue	Wed	Thu	Fri
v38	Project start (C015 and C016, ALC) →	→	→	→	Data collection (M209)
v39	→	→	Data analysis (M209)		
v40					
v41	Prototypes (C015 and C016, ALC)			Usability tests (M209)	
v42					
v43					
v44			Presentations	Presentations	

Analysis

Analysis means that you take something down into smaller parts and with the help of the parts you try to understand the whole.



Design Process



Preece, J., Sharp, H., & Rogers, Y. (2015). **Interaction design: beyond human-computer interaction**. John Wiley & Sons.

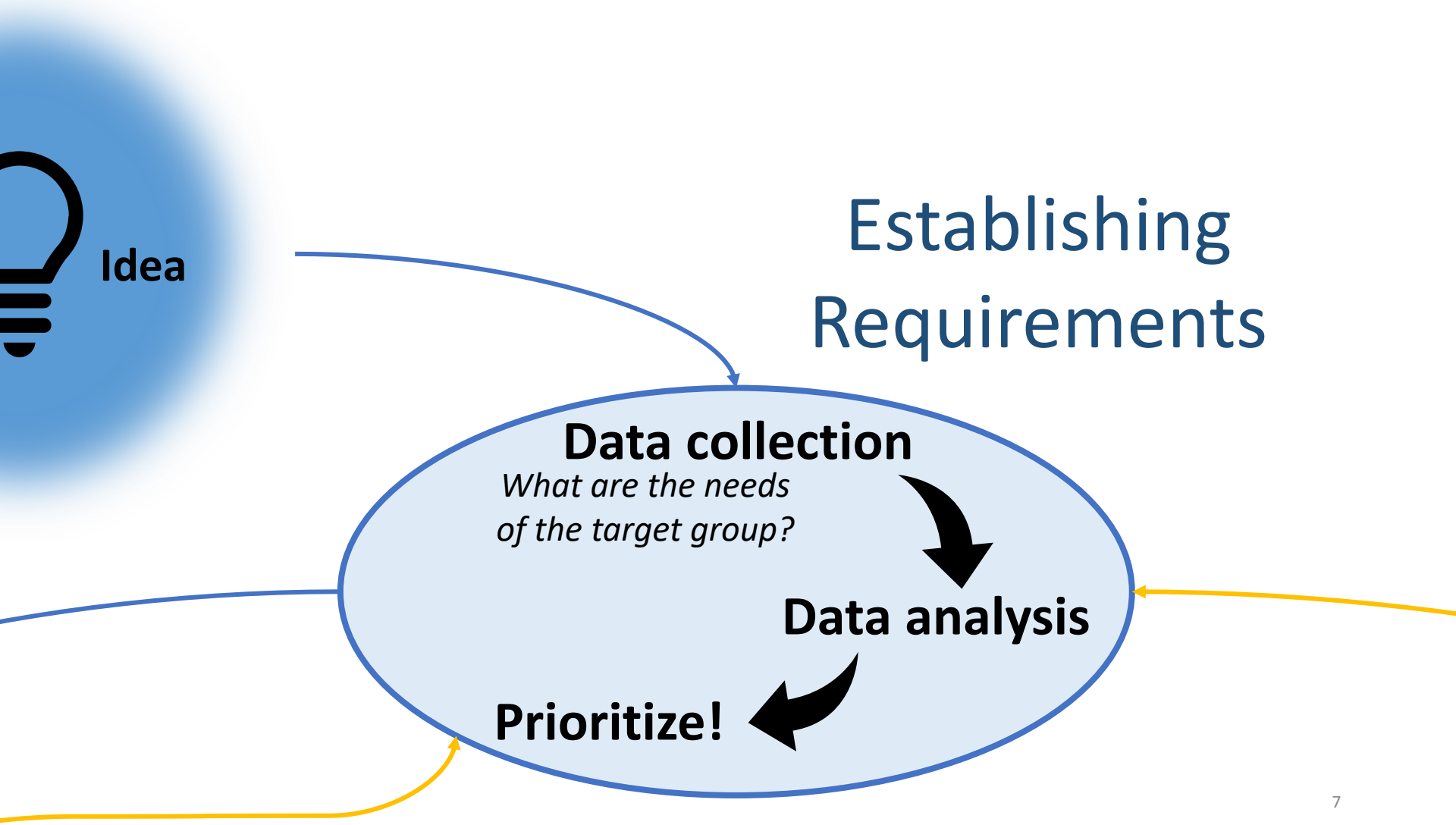
Design Process



Idea

Establishing Requirements

Design Process

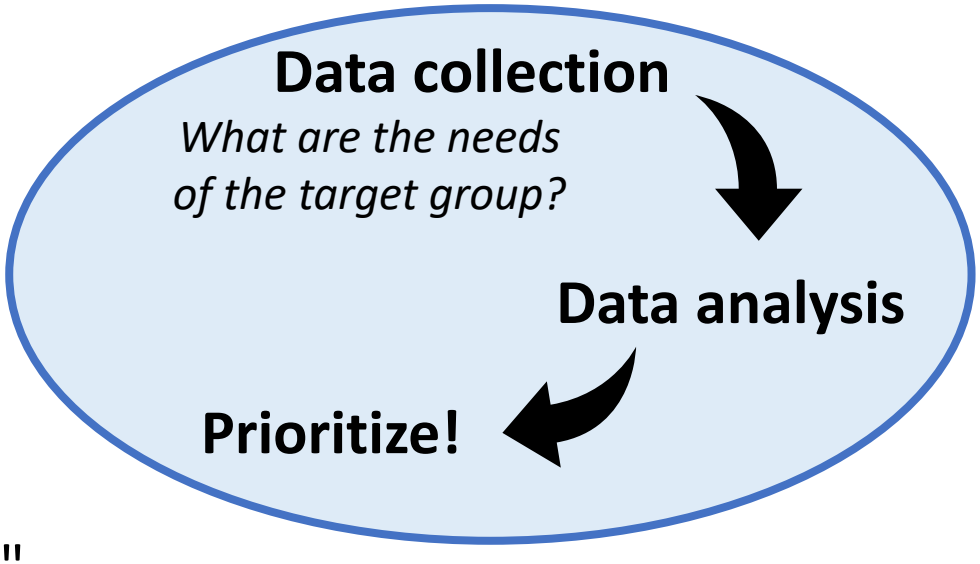


First Stage – Establish requirements

Find out if your target group has the need/problem you think

Through following steps:

- Data collection
- Data analysis
- Possibly update target group and/or needs/problems



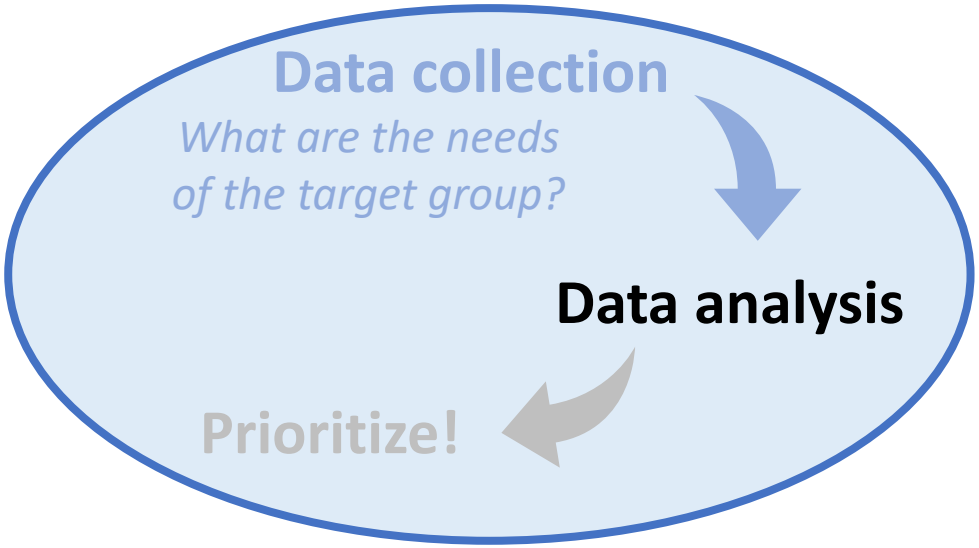
- Produce a "requirements list" (prioritized list of features)

The next step: Analysis!

What to do with the data?

What type of data have you gathered?

What to do next?



Analysis

Quantitative data

Quantitative analysis

Data:

Numerical

Analysis:

- Graphical presentation makes it easier to identify patterns and get an overall picture.
- Statistical analysis.

Qualitative data

Qualitative analysis

Data:

Notes from interviews, quotes, pictures, films, etc.

Analysis:

- Identify (recurring) patterns or themes.
- Categorize
- Identify users (personas) and scenarios

Where do you start?

	Usual raw data	Example qualitative data	Example quantitative data	Initial processing steps
Interviews	Audio recordings. Interviewer notes. Video recordings	Responses to open questions. Video pictures. Respondent's opinions	Age, job role, years of experience. Responses to closed questions	Transcription of recordings. Expansion of notes
Questionnaires	Written responses. Online database	Responses to open questions. Responses in 'further comments' fields. Respondent's opinions	Age, job role, years of experience. Responses to closed questions	Clean up data. Filter into different data sets
Observation	Observer's notes. Photographs. Audio and video recordings. Data logs. Think-aloud	Records of behavior. Description of a task as it is undertaken. Copies of informal procedures	Demographics of participants. Time spent on a task. The number of people involved in an activity	Expansion of notes. Transcription of recordings. Synchronization between data recordings

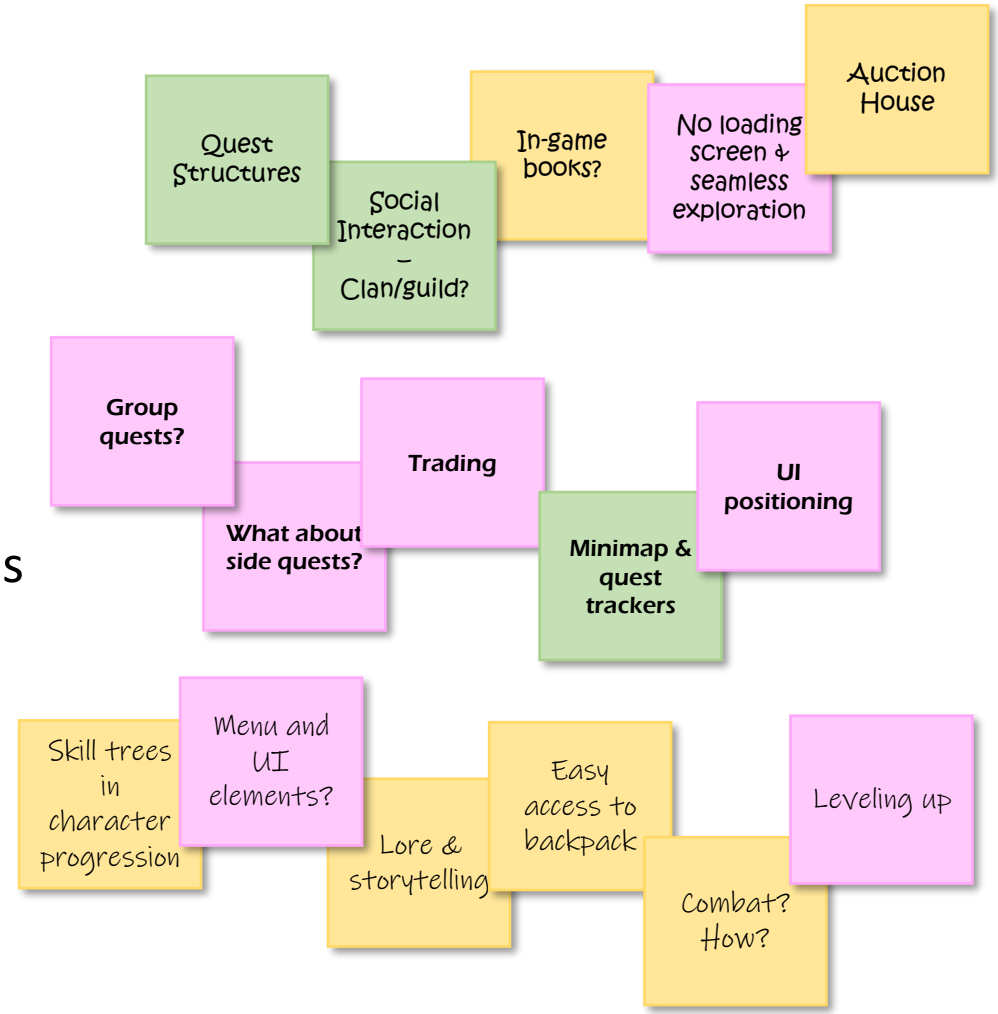
Table 8.1 Data gathered and typical initial processing steps for the main data gathering techniques

Example qualitative analysis
Affinity diagram

Structure and group the information.

A few steps:

- Record ideas.
- Look which ones are related.
- Sort them!

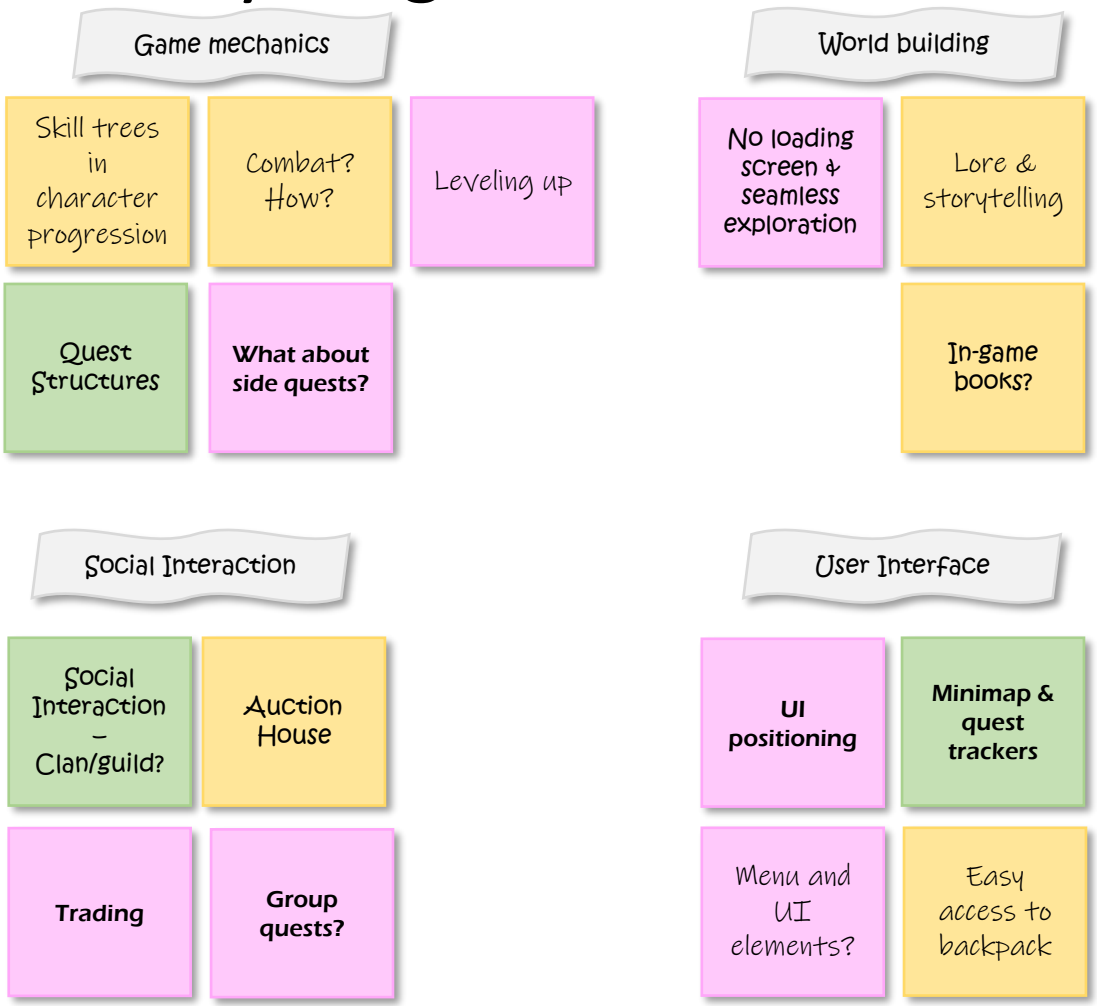


Example qualitative analysis
Affinity diagram

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A few steps:

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Example qualitative analysis
Affinity diagram

For example, everyone in the group reads notes from the interviews and marks interesting observations or ideas and writes these on different pieces of paper. Place all the pieces randomly on a board and try to group and find patterns.

Ask questions like:

- What was surprising?
- What did we expect to find that we didn't?
- What does the data tell us that we didn't know before?
- What in data makes us see things in a new way?

Task analysis / Effect map / Impact map

(describe what the user is doing)

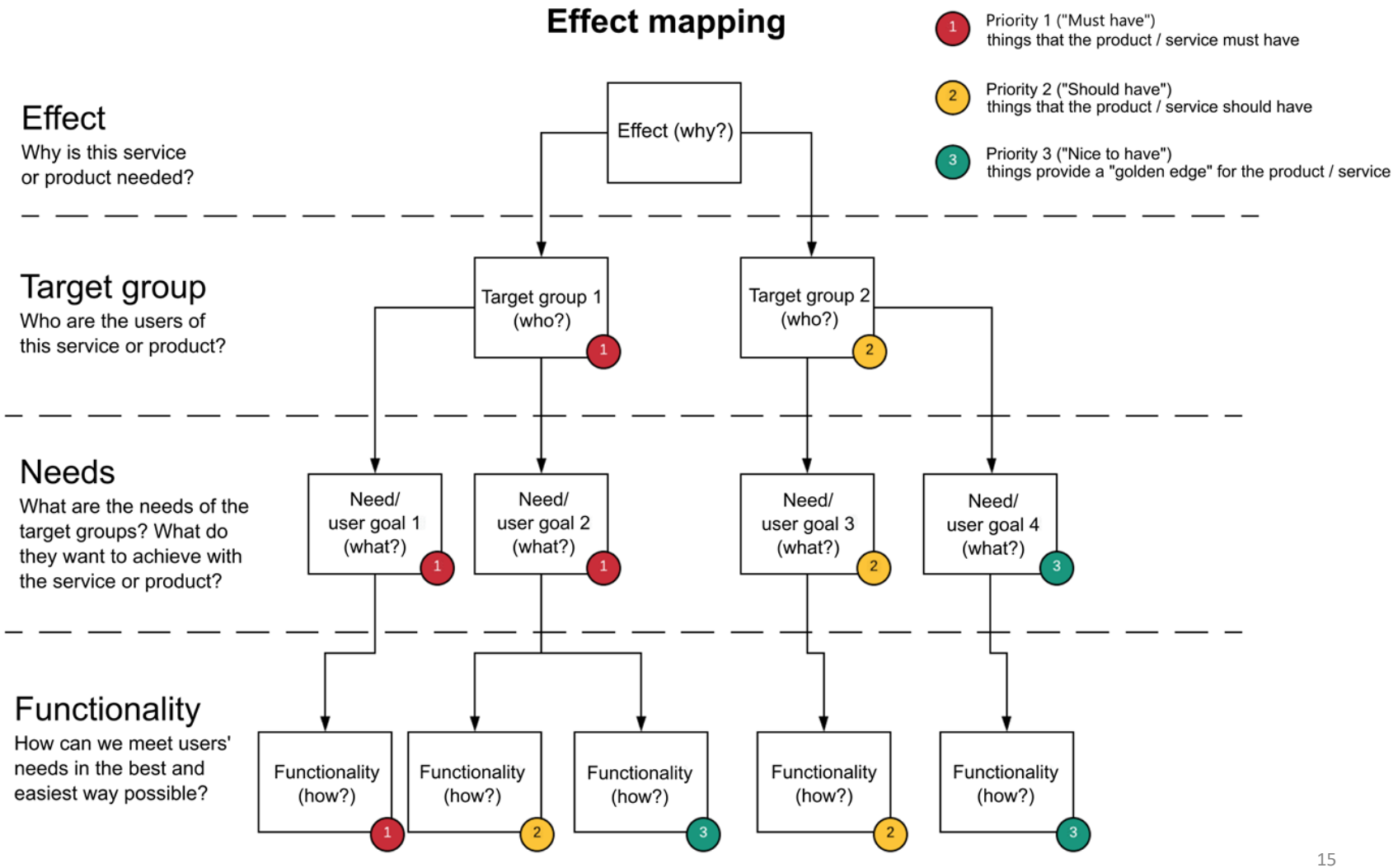


Image source: https://andreasjohanssonux.se/effektkarta/

Personas

Persona:	USDA Senior Manager Gatekeeper
Photo:	
Fictional name:	Matthew Johnson
Job title/ major responsibilities:	Program Staff Director, USDA
Demographics:	• 51 years old • Married • Father of three children • Grandfather of one child • Has a Ph.D. in Agricultural Economics.
Goals and tasks:	He is focused, goal-oriented within a strong leadership role. One of his concerns is maintaining quality across all output of programs. Spends his work time: • Requesting and reviewing research reports, • preparing memos and briefs for agency heads, and • supervising staff efforts in food safety and inspection.
Environment:	He is comfortable using a computer and refers to himself as an intermediate Internet user. He is connected via a T1 connection at work and dial-up at home. He uses email extensively and uses the web about 1.5 hours during his work day.
Quote:	“Can you get me that staff analysis by Tuesday?”

Source: [Personas @usability.gov]

Namn

Min motivation

Mina användarmål

Mina problemområden

Citat

Ålder:

Relation:

Arbete:

Boende:

Transport:

Ekonomi:

	Funktioner	Innehåll
Exiters Delighters		
Bra att ha		
Måste ha		

Egenskap:

Låg — Hög

Egenskap:

Låg — Hög

Egenskap:

Låg — Hög

Vår största utmaning

Persona template

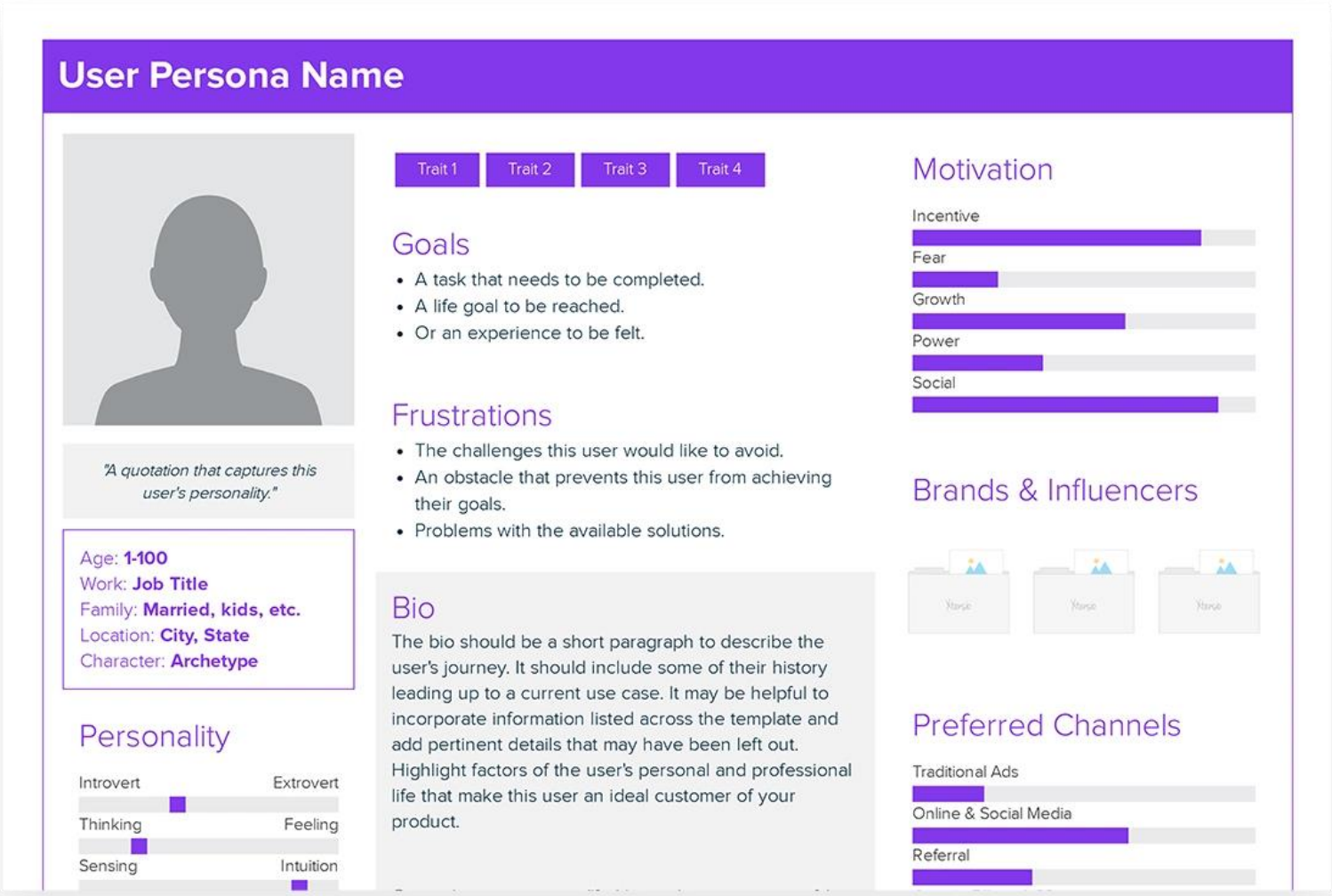


Image source: Xtensio

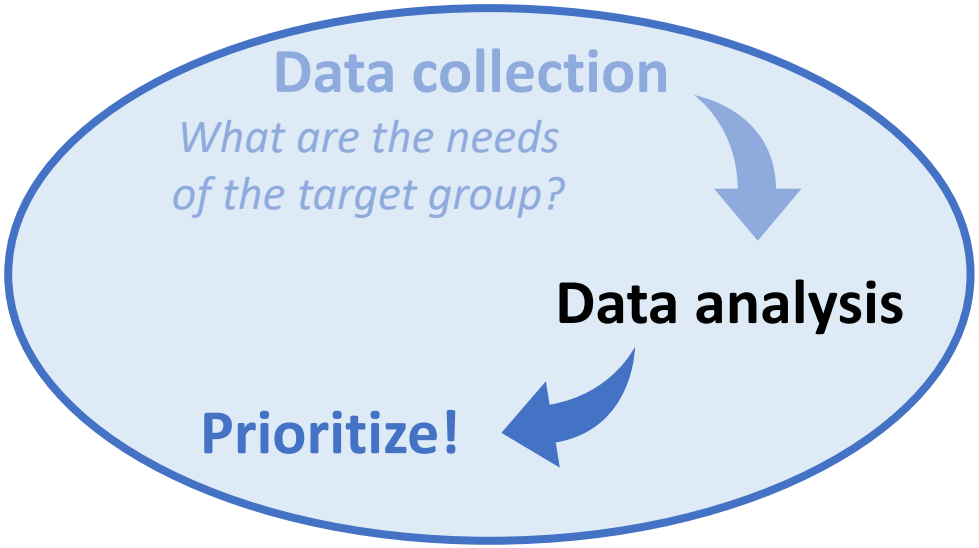
List of Requirements

Now combine

- what you know about content,
- functions
- usability goals,
- user experience goals, etc.

in a ***list of requirements***.

Prioritize!



You will need to focus on a few features as you work on your interaction design.

Exercise – Data analysis

- If you already have some data!

After data collection:

- **Select** analysis method(s).
- **Analyze** generate insight from data
- **Develop** use cases, scenarios, personas.
- **List** functions to be in your design.
- **Prioritize**
- **Focus** on the most important functions.

*We go through **prototyping** the next exercise.*

- Continue working on ideation and data collection

Create one or two personas, depending on your target group(s).